

1. Administrative & Study Identification

Full Study Title: Frequency of Selected Single Nucleotide Polymorphisms in Phase With the Mutant and Wild-Type HTT Alleles in Huntington Disease Gene Expansion Carriers **Short Title/ Acronym:** Frequency of Selected SNPs in HD Gene Expansion Carriers **Protocol Number:** WE45491 **NCT Number:** NCT06667414 **Sponsor Name:** Hoffmann-La Roche **Investigational Product:** N/A (Observational study; Blood sample for sequencing assay) **Phase of Development:** Phase N/A (Epidemiological / Observational) **Medical Monitor:** Hoffmann-La Roche Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To find out the frequency and types of genetic differences, called single nucleotide polymorphisms (SNPs), present in the wild-type Huntington (wtHTT) and mutant Huntington (mHTT) alleles. **Secondary Objectives:** To determine if these genetic differences are more common in some parts of the world based on demographics including sex, age, race, and ethnicity.

2.2 Background & Rationale Huntington's disease (HD) is a genetic disease caused by a mutated version of the huntingtin (HTT) gene. There is a high medical need to develop therapies that treat HD. Understanding more about genetic variations (SNPs) involved in HD may help drug developers direct research efforts. SNPs can potentially be used as an optimization target (SNP phasing identification for future allele-selective therapies) to guide selective investigational medicines.

3. Study Design & Methodology

3.1 Design Framework Study Type: Observational / Epidemiological **Medical Methodology:** Epidemiological genetic screening via blood draw **Control Type:** Single-arm (No control) **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration Target \$N\$: Recruitment ongoing across ~30 sites worldwide **Number of Sites:** Global (over 30 sites worldwide) **Estimated Study Start:** 02-Sep-2024 **Estimated Primary Completion:** 24-Sep-2028

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Aged 25 to 60 years, inclusive, at the time of signing the Informed Consent Form (ICF). 2. Primary Condition and Diagnosis Threshold: Confirmation of Huntington Disease (HD) gene expansion mutation carrier status. 3. Confirmation of Total Functional Capacity (TFC) ≥ 9 and Total Motor Score (TMS) > 6 within 12 months prior to signing the ICF; Ability to tolerate blood draws.

4.2 Exclusion Criteria 1. None listed for this observational study.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: N/A. Single-day visit for a one-time blood sample collection. **Management Pathway:** Single-day clinic visit protocol. **Route of Administration:** Intravenous (blood draw). **Duration of Treatment:** Single-day visit.

5.2 Concomitant & Prohibited Therapy Permitted: Standard of care medical treatments and medications. **Prohibited:** None specified.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Visit 1	Day 1 Informed Consent, Demographics Questionnaire, Medical & Medication History Review, Blood Draw for SNP Sequencing Assay

Onsite Visit Cadence: Day 1 **Total Duration:** Single-day visit

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Frequency of selected single nucleotide polymorphisms (SNPs) phased on the wild-type Huntington (wtHTT) and mutant Huntington (mHTT) alleles. **Measurement Method:** Blood sample testing via a specifically designed central laboratory sequencing assay.

7.2 Secondary & Exploratory Endpoints Distribution of genetic variations across different geographic, racial, and ethnic demographics.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Patient monitoring for unwanted effects of the blood sample collection (e.g., pain, bruising,

infection, dizziness, fainting, or upset stomach). **Serious Adverse Events (SAEs):** Standard reporting timeline for observational and epidemiological trials. **Data Monitoring Committee (DMC):** Internal sponsor review.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: All enrolled participants providing a viable blood sample for the sequencing assay. **Sample Size Justification:** Epidemiological estimation; powered to identify significant global and regional frequencies of targeted HTT SNPs. **Handling of Missing Data:** Exclusion of missing or unviable genetic samples from the primary SNP frequency calculation.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine verification of informed consent (ICF) and sample integrity. **Remote Monitoring Frequency:** N/A **Checklist Review Interval:** N/A **Quality Audit Mechanism:** Central laboratory genetic sequencing assay validation. **Audit Trail:** eCRF tracking for demographics, functional scores, and sample chain of custody. **Training & Education Protocols:** - **HCP:** Site initiation and protocol training. - **Patient:** Informed consent and study objective education.

11. Study Identifiers & Governance

Primary ID: {{WE45491}} **Registry Number:** {{NCT06667414}} **Sponsor/Lead Organization:** {{Hoffmann-La Roche}} **Study Title:** {{Frequency of Selected SNPs in HD Gene Expansion Carriers}} **Official Regulatory Title:** {{Frequency of Selected Single Nucleotide Polymorphisms in Phase With the Mutant and Wild-Type HTT Alleles in Huntington Disease Gene Expansion Carriers}}

12. Clinical Framework (Huntington Disease Specific)

Primary Condition: {{Huntington Disease}} **Diagnosis Threshold:** {{HD gene expansion mutation carrier status, TFC ≥ 9 , TMS > 6 }} **Medical Methodology:** {{Epidemiological genetic screening via blood draw}} **Optimization Target:** {{SNP phasing identification for future allele-selective therapies}}

13. Study Procedures & Methodology

Management Pathway: {{Single-day clinic visit protocol}}
Education Protocol (HCP): {{Site initiation and protocol training}} **Education Protocol (Patient):** {{Informed consent and study objective education}} **Quality Audit Mechanism:** {{Central laboratory genetic sequencing assay validation}}

14. Observation & Follow-Up Schedule

Total Duration: {{Single-day visit}} **Onsite Visit Cadence:** {{Day 1}} **Remote Monitoring Frequency:** {{N/A}} **Checklist Review Interval:** {{N/A}}

15. Sign-off & Approval

Role Name	Signature	Date	:	---	:	---	:	---	:
Principal Investigator	_____	[DD-MMM-YYYY]	:		:		:		
Operations Lead	_____	[DD-MMM-YYYY]	:		:		:		
	_____	[DD-MMM-YYYY]	:		:		:		